

Module 3 | NumPy, Pandas & Matplotlib

Technical Workouts

1. Understand NumPy Fundamentals
 - a. Why NumPy is used
 - b. Arrays vs Python lists
 - c. Creating NumPy arrays
 - d. Array operations
 - e. Vectorized computation
2. Understand Numerical Computing
 - a. Mathematical operations
 - b. Broadcasting
 - c. Array indexing
 - d. Array slicing
 - e. Multi-dimensional arrays
3. Understand Pandas Fundamentals
 - a. Series
 - b. DataFrames
 - c. Loading datasets
 - d. Inspecting data
 - e. Data exploration
4. Understand Data Manipulation
 - a. Filtering data
 - b. Sorting data
 - c. Grouping data
 - d. Aggregations
 - e. Handling missing values
5. Understand Data Analysis Workflows
 - a. Dataset inspection
 - b. Data summarization
 - c. Data transformations
 - d. Feature exploration
 - e. Analytical thinking

6. Understand Matplotlib Fundamentals

- a. Why visualization matters
- b. Plot creation
- c. Labels and legends
- d. Chart customization
- e. Visual storytelling

7. Understand Common Visualizations

- a. Line charts
- b. Bar charts
- c. Histograms
- d. Scatter plots
- e. Distribution visualization

8. Thinking with AI

- a. Use AI for:
 - i. Data analysis suggestions
 - ii. Visualization recommendations
 - iii. Dataset interpretation
- b. Validate generated insights
- c. Verify visual conclusions

9. Apply Through Hands-on Tasks

- a. Analyze real-world datasets
- b. Create visualizations
- c. Perform data transformations
- d. Generate summary reports
- e. Explore relationships in data
- f. Present findings using charts